

CORRECTIONS/ERRATA – MATH

Item 1

Question

Correct Answer

If the function $y = 2x^2$ is integrated between the limits of $x = -3$ and $x = +3$, the result is ____.

54

Solution:

$$\int_{-3}^3 2x^2 dx = \frac{2}{3} x^3 \Big|_{-3}^3 = 54 \rightarrow \text{Ans}$$

The correct answer is 36

Question

Correct Answer

The integral of $\sec^2 x \tan^3 x$ dx from 0 to $\pi/3$ equals ____.

(Sqrt 3)/3

Solution:

$$\int_0^{\pi/3} \sec^2(x) \tan^3(x) dx = \int_0^{\pi/3} (\tan x)^2 (\sec^2 x dx)$$
$$= \frac{\tan^3(x)}{3} \Big|_0^{\pi/3}$$

The correct answer is sqrt 3

41

Question

Correct Answer

Four is added to the quantity two minus the sum of negative seven and six. This answer is then multiplied by three. What is the result?

-21

Solution:

$$x = 3 \left(2 - (-7 + 6) \right) + 4$$
$$x = -21 \rightarrow \text{Ans}$$

The correct answer is 21

Question

Correct Answer

When a metallic ball bearing is placed inside a cylindrical container of radius 2 cm, the height of water inside the cylinder increases by 0.6 cm. What is the radius of the ball bearing?

1.8 cm

Solution:

$$V_{\text{ball}} = \pi r_{\text{cyl}}^2 (\Delta h)$$
$$\frac{4}{3} \pi r^3 = \pi (2)^2 (0.6)$$
$$r = 1.8 \text{ cm} \rightarrow \text{Ans}$$

The correct answer is 1.2

Question

Correct Answer

Evaluate the integral of $\sqrt{1 - 2 \cos \alpha}$ dα with limits from 0 to $\pi/4$.

Solution:

Using calculator:

$$\int_0^{\pi/4} \sqrt{1 - 2 \cos \alpha} d\alpha = 0.414 \rightarrow \text{Ans}$$

Correct answer is 0.414. Given should be (1 – cos 2α)

Question

Correct Answer

When partial fractions are used, the decomposition of $(x - 1)/(x^2 + 3x + 2)$ is equal to:

$-2/(x + 1) + 3/(x + 2)$

Correct answer is -2/(x + 1) + 3/(x + 2)

Question

Correct Answer

A parabola has its focus at (7,-4) and directrix y = 2. Find its equation.

Solution:

By inspection, $2a = 6$ and $a = 3$
The coordinates of the vertex is at (7,-1).
Standard equation: $(x - h)^2 = -4a(y - k)$
Substituting:
 $(x - 7)^2 = -4(3)(y + 1)$

The correct answer is $x^2 - 14x + 12y + 61 = 0$

Question

Correct Answer

Rabbits in a lab are to be kept on a strict daily diet that includes 30 grams of protein, 16 grams of fat and 24 grams of carbohydrates. The scientist has only three food mixes available with the following grams of nutrients per unit.

Mixture	Protein	Fat	Carbohydrates
Mix A	4	6	3
Mix B	6	1	2
Mix C	4	1	12

The question is “Find how many units of mix B are needed daily to meet each rabbits dietary need.”

Question

Correct Answer

Find the slope of the line whose parametric equations are $x = 4t + 6$ and $y = t - 1$.

$y' = y/2x$

Solution:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{1}{4} \rightarrow \text{Ans}$$

Correct answer is 1/4

Question

Correct Answer

Evaluate $\int \frac{\ln(2x - 1)^2}{4(2x + 1)} dx$.

$\ln(2x - 1)^2 + C$

$\ln(2x - 1) + C$

Solution:

Correct solution shown below

$$= \int \frac{4}{2x + 1} dx$$

Let,

$$u = 2x + 1$$
$$du = 2dx$$
$$= \int \frac{2}{u} du$$
$$= 2 \ln u + C$$
$$= 2 \ln(2x + 1) + C \rightarrow \text{Ans}$$

Question

Correct Answer

The volume of the solid generated by revolving the area bounded by $y = \sqrt{x}$, $y = 2$, and the y - axis about the y - axis is:

A. 8π
B. $16\pi/3$
C. $32\pi/3$ - Ans
D. $8\pi/3$

Solution:

$$y_U = 2 \quad y_L = \sqrt{x}$$

Solving intersection:
 $y_U = y_L$
 $2 = \sqrt{x} \Rightarrow x = 4$
and $x = 0$ (y - axis)

Using Pappus' Theorem, $V = \int 2\pi R dA$
where, $R = x$
 $dA = (y_U - y_L) dx = (2 - \sqrt{x}) dx$
 $V = \int_0^4 2\pi x (2 - \sqrt{x}) dx = \frac{32\pi}{5} \rightarrow \text{Ans}$

Question

Correct Answer

The polishing time for a bowl is 6, 12 minutes. What is the average polishing time?

A. 2.9076
B. 9.6621
Solution:
 $\bar{X} = \frac{6 + 12 + 1}{3} = 7.6667$

Correct answer

Correct answer 32pi/5

Determine the initial value of the time-domain response of the following equation using the initial-value theorem. $Y(s) = 2s/(s + 1)^2(s + 2)$.

Correct answer is 0. Solution shown below

$$Y(s) = \frac{2s}{(s+1)^2(s+2)}$$

For initial value,

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s)$$

$$= \lim_{s \rightarrow \infty} s \frac{2s}{(s+1)^2(s+2)}$$

$$= \lim_{s \rightarrow \infty} \frac{2s^2}{s^3 + 4s^2 + 5s + 2}$$

$$= 0 \rightarrow \text{Ans}$$